

PH101: 2022

Quantum Mechanics

Lecture 7

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Recap

- Separable solutions provide energy eigenstates, However for free particle, there is no such case, they can only be represented by wave packet.
- Solutions are classified as Bound state and / or Scattering state, depending on the kind of potential that particle is subject to.
 - Step potential provides scattering state solution. For $E < V_0$
For step potential we find the reflection coefficient $R = 1$
- The wave function penetrates into the barrier, classically forbidden region even if the energy is smaller than potential i.e. $E < V_0$ - Barrier Penetration.

Energy is greater than step height

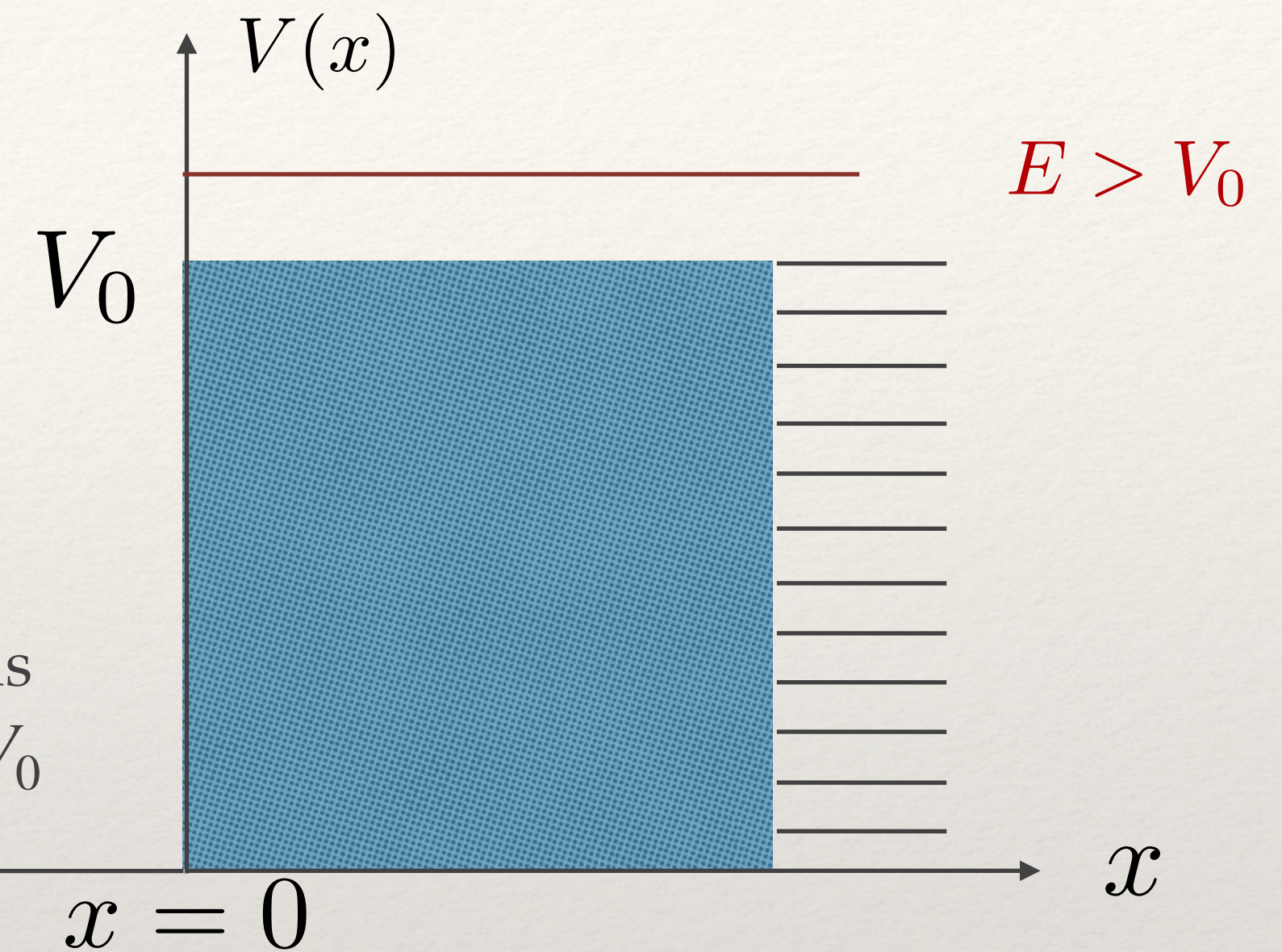
$$E > V_0$$

Time independent Schrodinger Equation:

$$x < 0 : \frac{d^2\psi}{dx^2} + k_1^2\psi = 0; \quad k_1 = \left(\frac{2mE}{\hbar^2}\right)^{1/2}$$

$$x > 0 : \frac{d^2\psi}{dx^2} + k_2^2\psi = 0; \quad k_2 = \left[\frac{2m}{\hbar^2}(E - V_0)\right]^{1/2}$$

Real as $E > V_0$
This equation was different for $E < V_0$



Solution:

$$x < 0 : \psi(x) = Ae^{ik_1x} + Be^{-ik_1x}$$

$$x > 0 : \psi(x) = Ce^{ik_2x} + \cancel{De^{-ik_2x}}$$

This term provides the possibility of the wave travelling in the direction of **decreasing** x and we can omit it. As if transmitted part of the wave is travelling towards positive x.

Boundary Conditions

Continuity of the wave function and its first derivative at boundary

$$\left\{ \begin{array}{l} \psi_I(x)|_{x=0} = \psi_{II}(x)|_{x=0} \longrightarrow A + B = C \\ \frac{\partial\psi_I(x)}{\partial x}|_{x=0} = \frac{\partial\psi_{II}(x)}{\partial x}|_{x=0} \longrightarrow k_1(A - B) = k_2C \end{array} \right.$$

Solution

Simple algebra:

$$B = \frac{k_1 - k_2}{k_1 + k_2} A, \quad C = \frac{2k_1}{k_1 + k_2} A$$

Solution:

$$x < 0 : \quad \psi(x) = Ae^{ik_1x} + \frac{k_1 - k_2}{k_1 + k_2} Ae^{-ik_1x}$$

$$x > 0 : \quad \psi(x) = \frac{2k_1}{k_1 + k_2} Ae^{ik_2x}$$

• Reflection coefficient:

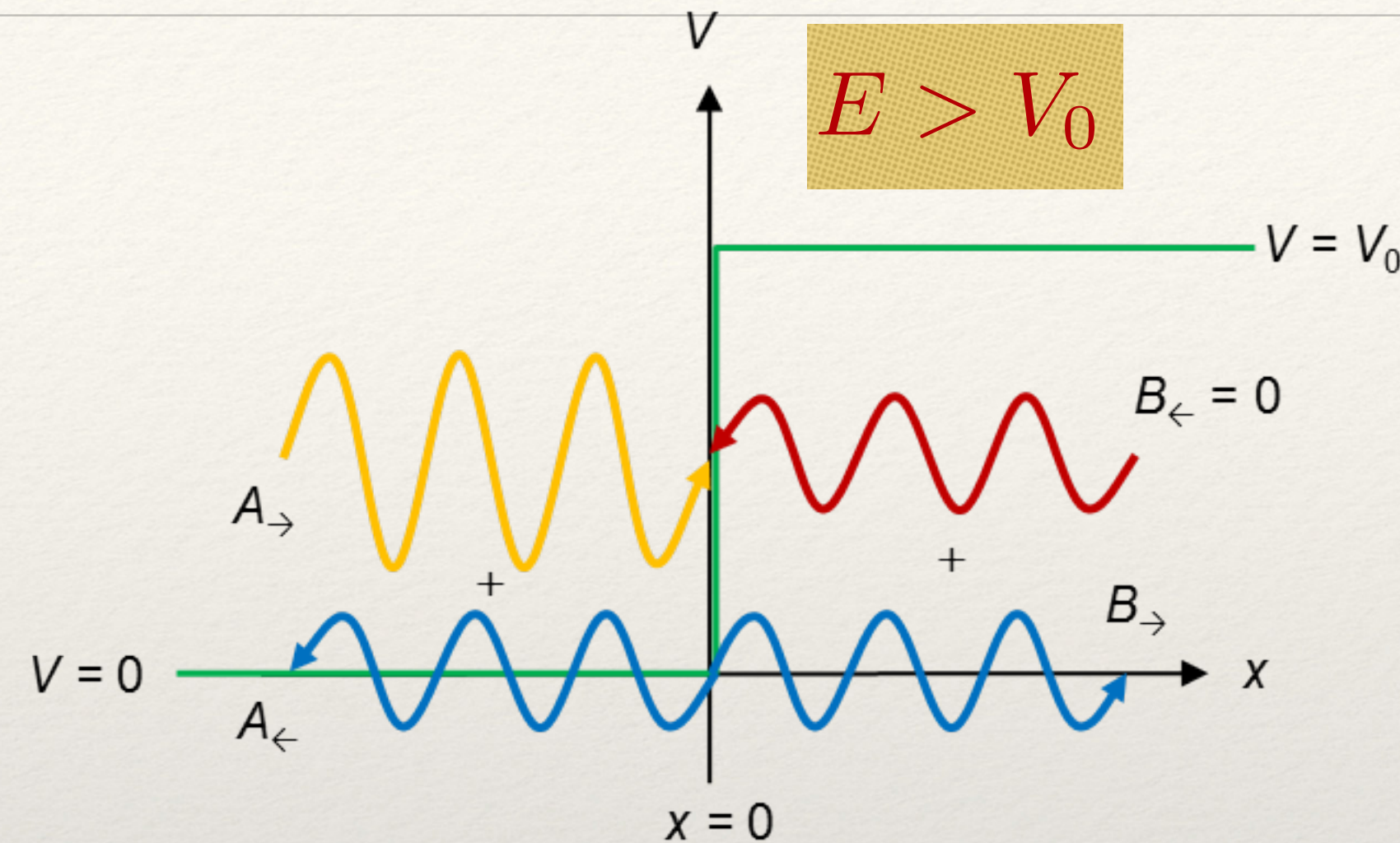
$$R = \frac{|B|^2}{|A|^2} = \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2 < 1$$

• Transmission coefficient:

$$T = \frac{v_2 |C|^2}{v_1 |A|^2}$$

Transmission or reflection coefficients are ratios of probability fluxes. For R, the velocity of the waves are same, while for transmission coefficient they are different. So, it needs to be accounted.

This is also in contrast with the classical solution, as with even higher energy, the particle can feel the potential



$$\begin{aligned} A_{\rightarrow} &= A \\ A_{\leftarrow} &= \frac{k_1 - k_2}{k_1 + k_2} A = B \\ B_{\rightarrow} &= \frac{2k_1}{k_1 + k_2} A = C \end{aligned}$$

The probability flux incident upon potential step splits into reflected and transmitted ones and adds to one.

$$v_1 = \frac{p_1}{m} = \frac{\hbar k_1}{m}; \quad v_2 = \frac{p_2}{m} = \frac{\hbar k_2}{m}$$

$$T = \frac{k_2}{k_1} \frac{(2k_1)^2}{(k_1 + k_2)^2}$$

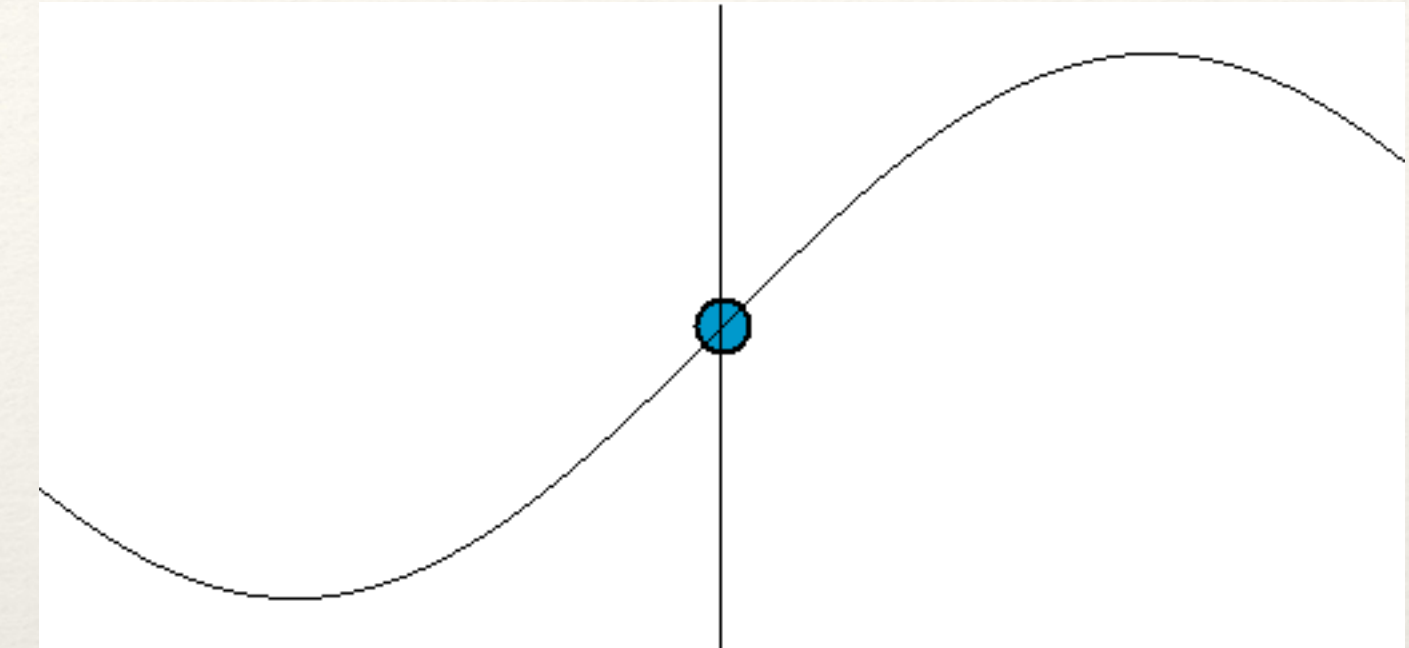
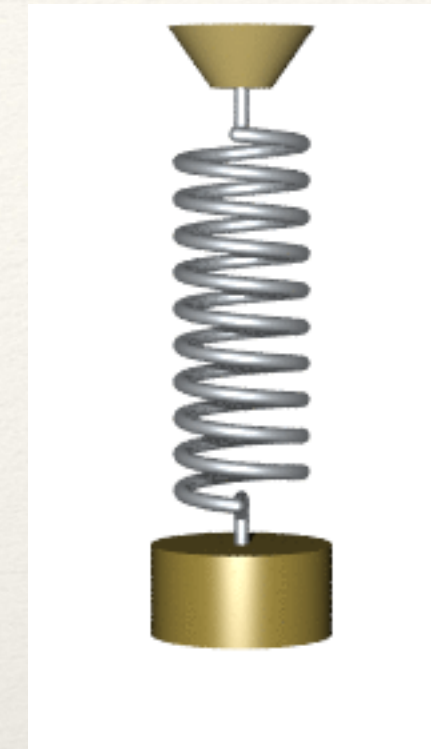
$$R + T = 1.$$

Harmonic Oscillator

Classical Equation
of motion:

$$\vec{F} = -k\vec{x} = m \frac{d^2x}{dt^2}$$

Example: Spring



Solution: $x(t) = A \sin(\omega t) + B \cos(\omega t),$

Simple Harmonic
motion

Frequency:

$$\omega \equiv \sqrt{\frac{k}{m}}$$

Time period
of oscillation:

$$T = 2\pi/\omega$$

Following the relation of force
to scalar potential :

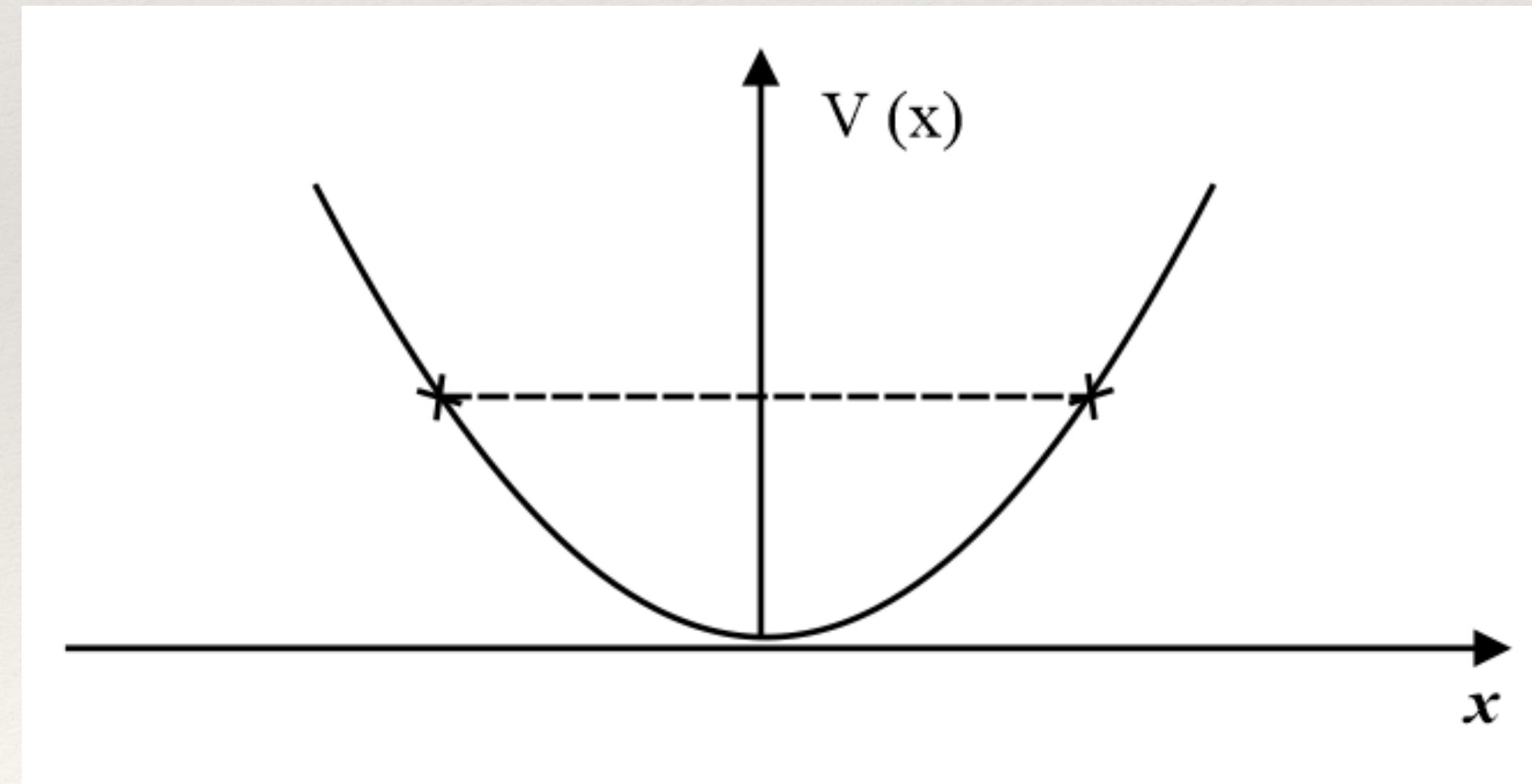
$$F = -kx = \frac{\partial V}{\partial x}$$

$$V = - \int \vec{F} \cdot d\vec{x} = \int kx \, dx$$

Unlike previous
examples, the
potential is
continuous in
space, more
realistic.

The
potential:

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2$$



Parabola

Symmetric: $V(-x) = V(x)$

Quantum Harmonic Oscillator

Consider a quantum particle in harmonic oscillator potential:

$$V(x) = \frac{1}{2}m\omega^2 x^2$$

Time independent Schrodinger Equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi = E\psi.$$

Even before solving, we can speculate what kind of solution we may obtain:

Bound state solution

There are two ways to solve it. One is **Algebraic method** and the other is **Analytic method**, by power series expansion.

Rewrite

Algebraic method

$$\frac{1}{2m} \left[\left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 + (m\omega x)^2 \right] \psi = E\psi.$$

So that it can be factorised into:

$$u^2 + v^2 = (u - iv)(u + iv).$$

Note however, u, v are not just functions, but they are operators that do not commute!

Lets define, two operators, exactly similar to u and v :

$$a_{\pm} \equiv \frac{1}{\sqrt{2m}} \left(\frac{\hbar}{i} \frac{d}{dx} \pm im\omega x \right).$$

Algebraic solution to Harmonic Oscillator

Check:

$$a_- a_+ = \frac{1}{2m} \left[\left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 + (m\omega x)^2 \right] + \frac{1}{2} \hbar \omega.$$

Time independent Schrodinger Equation:

$$(a_- a_+ - \frac{1}{2} \hbar \omega) \psi = E \psi.$$

One can similarly check that:

$$(a_+ a_- + \frac{1}{2} \hbar \omega) \psi = E \psi.$$

➔ $a_- a_+ - a_+ a_- = \hbar \omega,$ $[a_-, a_+] = \hbar \omega$

$$\begin{aligned} (a_- a_+) f(x) &= \frac{1}{2m} \left(\frac{\hbar}{i} \frac{d}{dx} - im\omega x \right) \left(\frac{\hbar}{i} \frac{d}{dx} + im\omega x \right) f(x) \\ &= \frac{1}{2m} \left(\frac{\hbar}{i} \frac{d}{dx} - im\omega x \right) \left(\frac{\hbar}{i} \frac{df}{dx} + im\omega x f \right) \\ &= \frac{1}{2m} \left[-\hbar^2 \frac{d^2 f}{dx^2} + \hbar m \omega \frac{d}{dx} (x f) - \hbar m \omega x \frac{df}{dx} + (m\omega x)^2 f \right] \\ &= \frac{1}{2m} \left[\left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 + (m\omega x)^2 + \hbar m \omega \right] f(x). \end{aligned}$$

A simple algebra as above yields,

$$a_+ a_- = \frac{1}{2m} \left[\left(\frac{\hbar}{i} \frac{d}{dx} \right)^2 + (m\omega x)^2 \right] - \frac{1}{2} \hbar \omega.$$

Some important properties of the operators

Claim:

If ψ satisfies Schrodinger Equation with energy eigenvalue E , then $a_+\psi$ satisfies the Equation with energy $(E + \hbar\omega)$.

Proof:

$$\begin{aligned}(a_+a_- + \frac{1}{2}\hbar\omega)(a_+\psi) &= (a_+a_-a_+ + \frac{1}{2}\hbar\omega a_+)\psi \\ &= a_+(a_-a_+ + \frac{1}{2}\hbar\omega)\psi = a_+[(a_-a_+ - \frac{1}{2}\hbar\omega)\psi + \hbar\omega\psi] \\ &= a_+(E\psi + \hbar\omega\psi) = (E + \hbar\omega)(a_+\psi).\end{aligned}$$

Claim:

Similarly, $a_-\psi$ satisfies the Schrodinger Equation with energy $(E - \hbar\omega)$.

Proof:

$$\begin{aligned}(a_-a_+ - \frac{1}{2}\hbar\omega)(a_-\psi) &= a_-(a_+a_- - \frac{1}{2}\hbar\omega)\psi \\ &= a_-[(a_+a_- + \frac{1}{2}\hbar\omega)\psi - \hbar\omega\psi] = a_-(E\psi - \hbar\omega\psi) \\ &= (E - \hbar\omega)(a_-\psi).\end{aligned}$$

Climbing up or down the ladder !

$a_{\pm}$ are called ladder operators.

a_{+} is a raising operator.

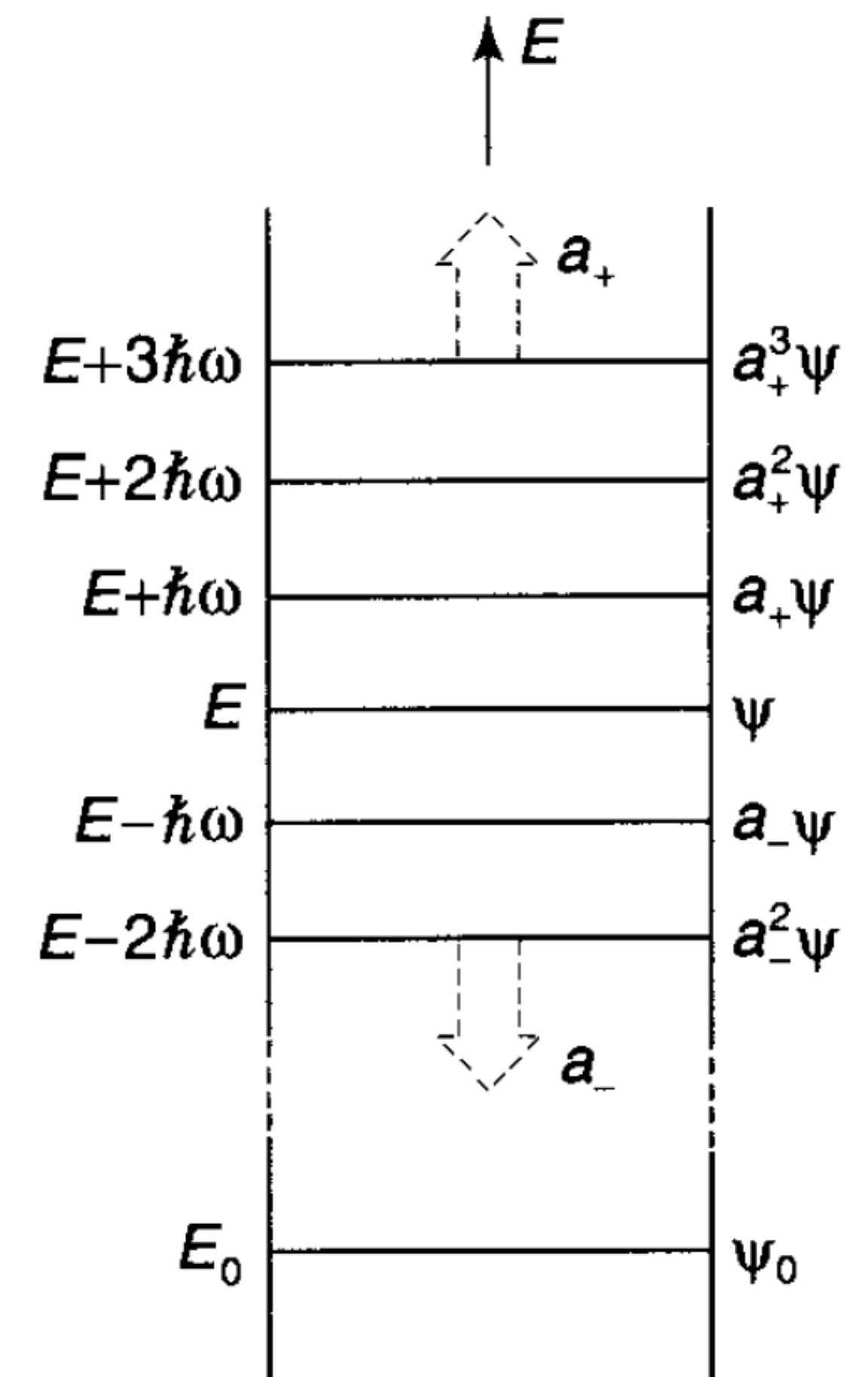
a_{-} is a lowering operator.

As they allow climbing
up and down the
energy eigenvalues

If we continue applying the lowering operator, then we will reach a state where applying it will lead to negative energy, which doesn't exist.

We can define that state to be the ground state of the system.

$$a_{-}\psi_0 = 0.$$



Ground state and excited states

$$a_- \psi_0 = 0.$$

Ground state is annihilated

$$\frac{1}{\sqrt{2m}} \left(\frac{\hbar}{i} \frac{d\psi_0}{dx} - im\omega x \psi_0 \right) = 0,$$

First order differential equation

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar} x \psi_0.$$

$$\int \frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar} \int x dx \Rightarrow \ln \psi_0 = -\frac{m\omega}{2\hbar} x^2 + \text{constant},$$

$$\psi_0(x) = A_0 e^{-\frac{m\omega}{2\hbar} x^2}.$$

Ground State wave function

Eigenvalue Equation for ground state

$$(a_+ a_- + (1/2)\hbar\omega) \psi_0 = E_0 \psi_0.$$

Following,

$$(a_+ a_- + \frac{1}{2}\hbar\omega) \psi = E \psi.$$

Now, $a_- \psi_0 = 0$

$$E_0 = \frac{1}{2}\hbar\omega.$$

Ground state energy

Sometimes called 'Zero point Energy'

Successive application of raising operator on the ground state yields the excited states:

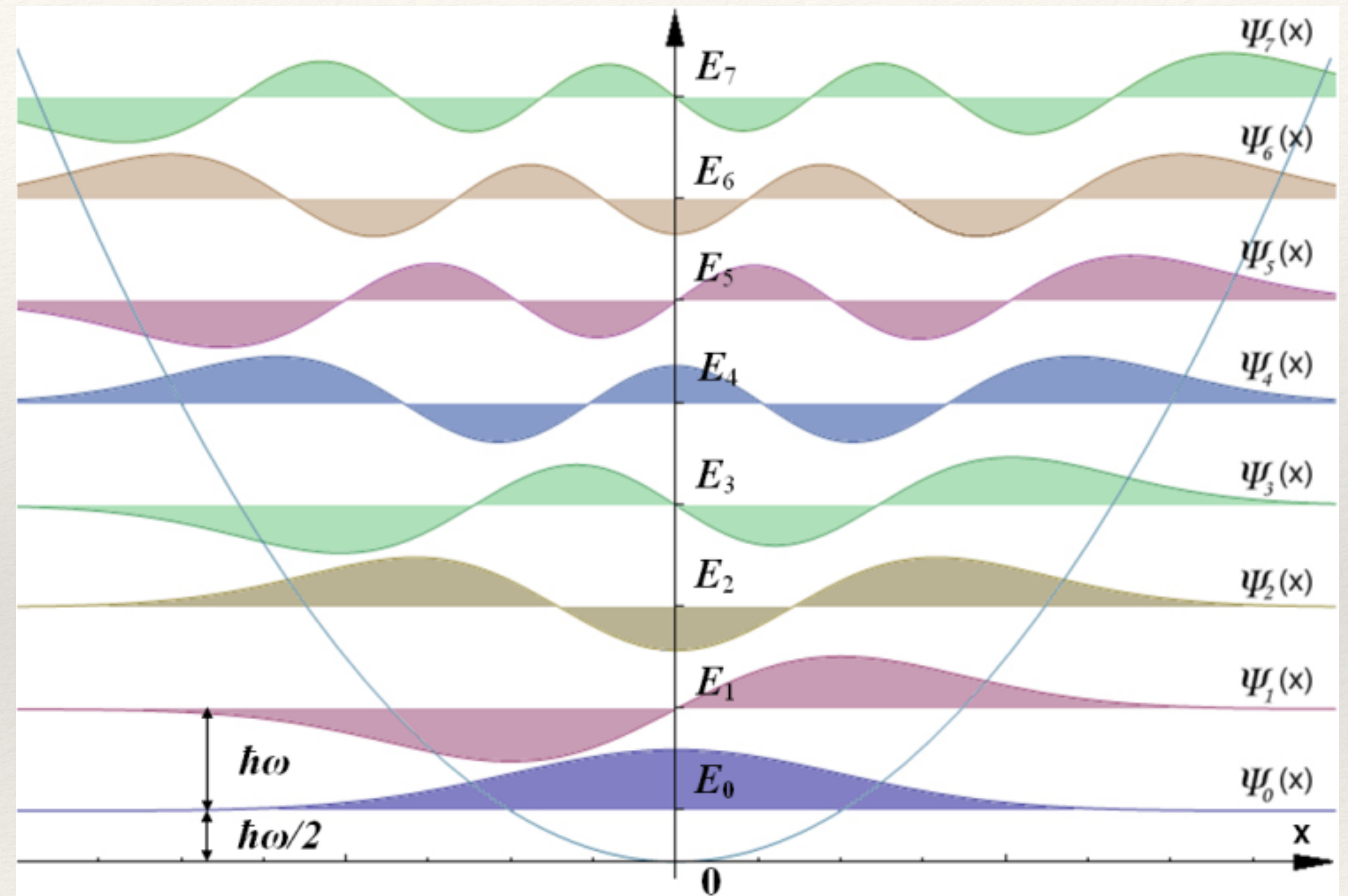
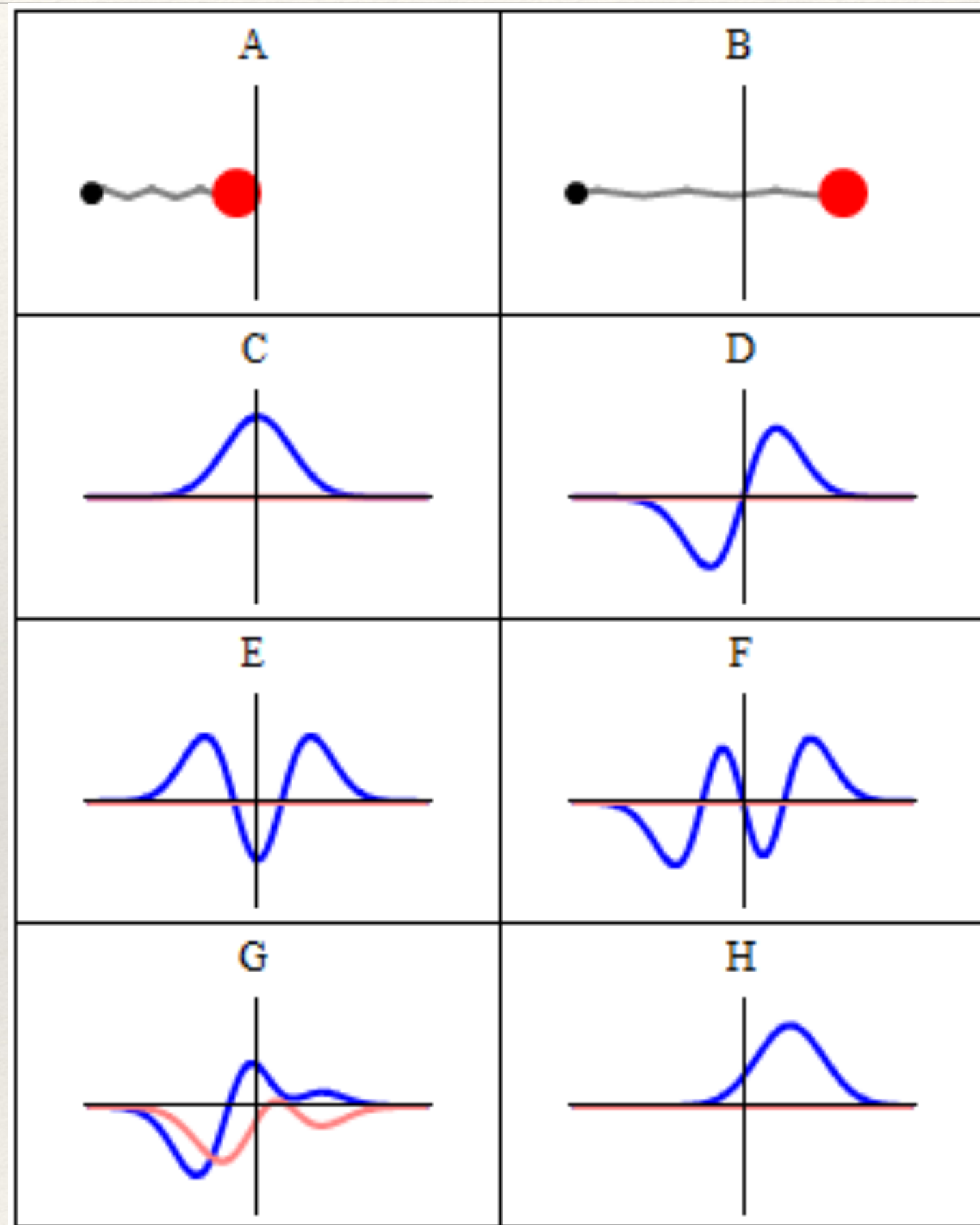
$$\psi_n(x) = A_n (a_+)^n e^{-\frac{m\omega}{2\hbar} x^2}, \quad \text{with } E_n = (n + \frac{1}{2})\hbar\omega. \\ n=0, 1, 2, \dots$$

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}.$$

Henkel function

We won't go into the details of it.

Eigenvalues and eigenstates for harmonic oscillator

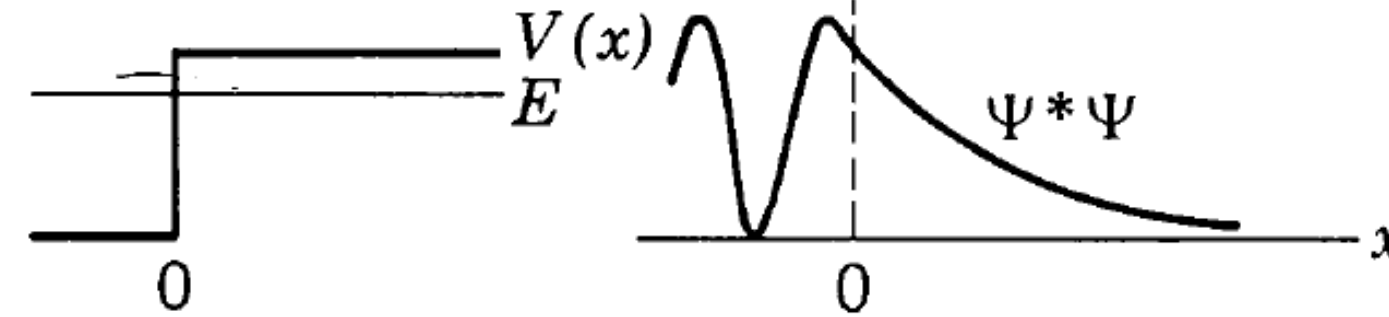


Trajectories in harmonic Oscillator; A, B: Classical,
C,D,E,F: Quantum Eigenstates (Blue: real, Red: imaginary)
G,H: Coherent states

First eight bound states of one dimensional quantum
harmonic oscillator

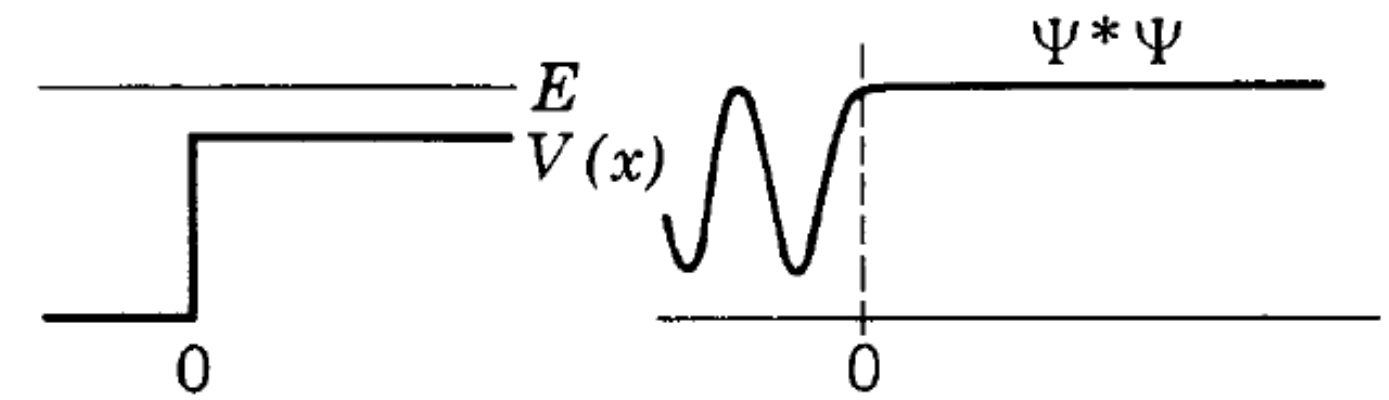
Some applications to physical cases

Step potential with energy below the top



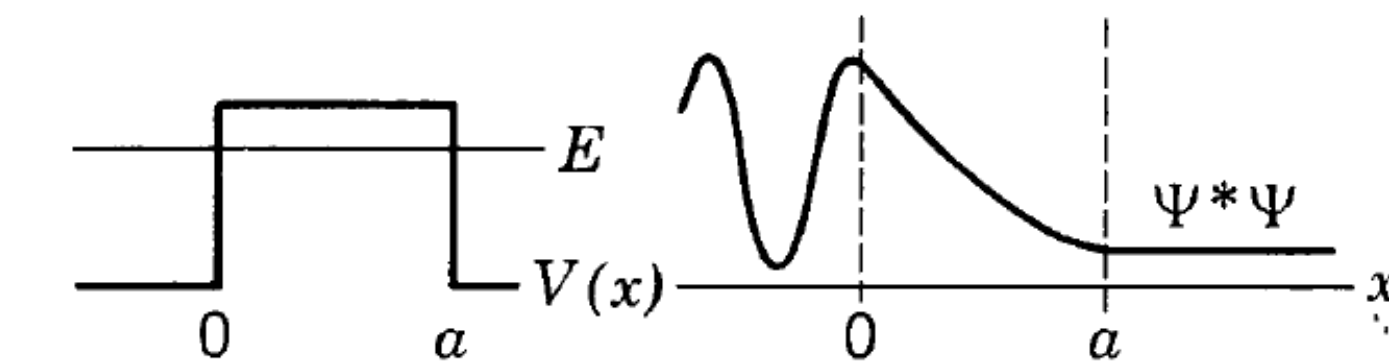
Conduction electrons near the surface of a metal

Step potential with energy above the top



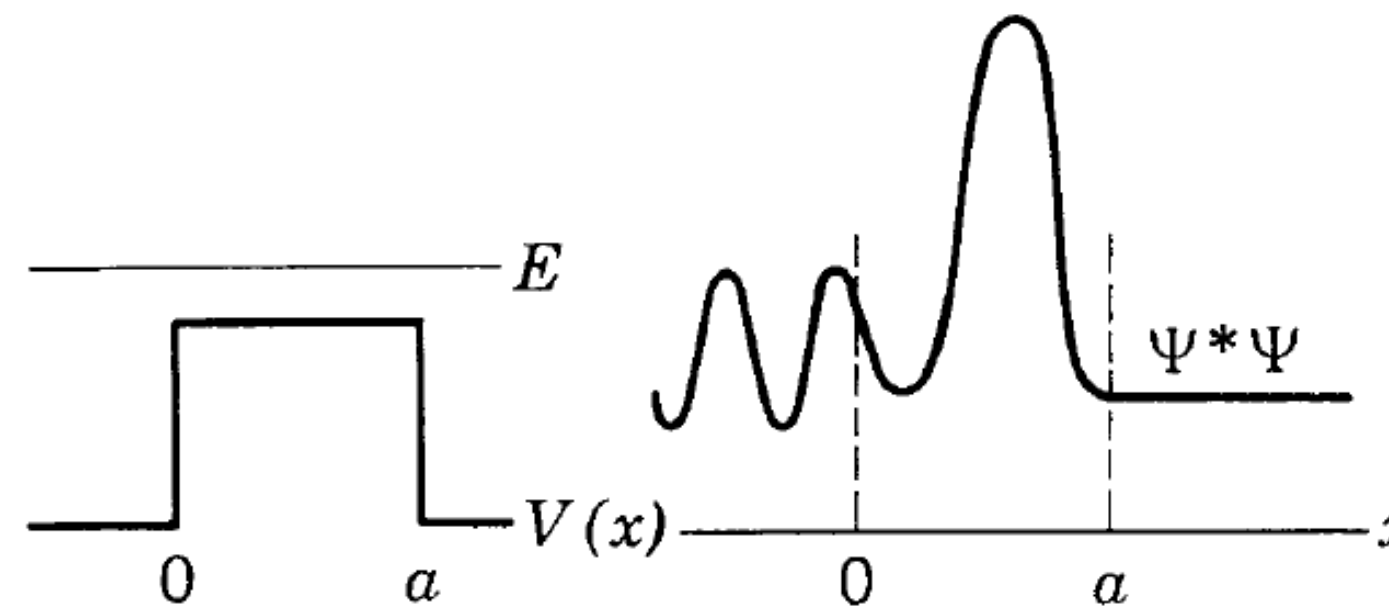
Neutrons trying to escape nucleus

Barrier potential with energy below the top



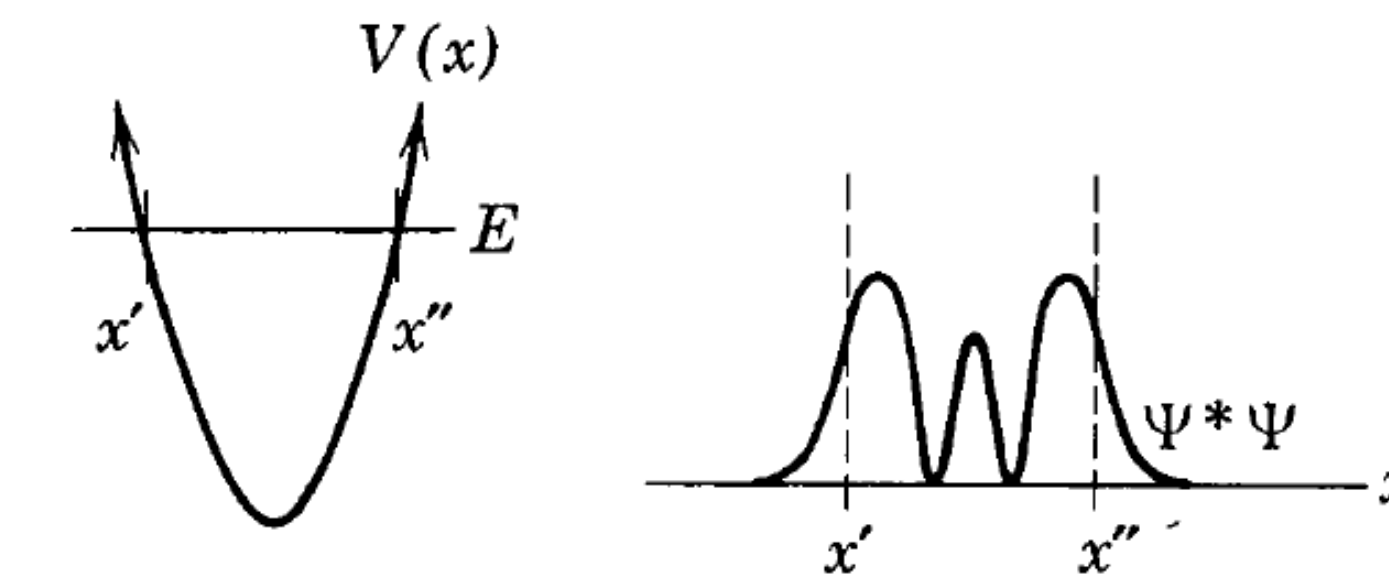
α Particles trying to escape Coulomb barrier

Barrier potential with energy above the top



Electron scattering from negatively charged atom

Simple harmonic oscillator potential



Atoms of vibrating diatomic molecules

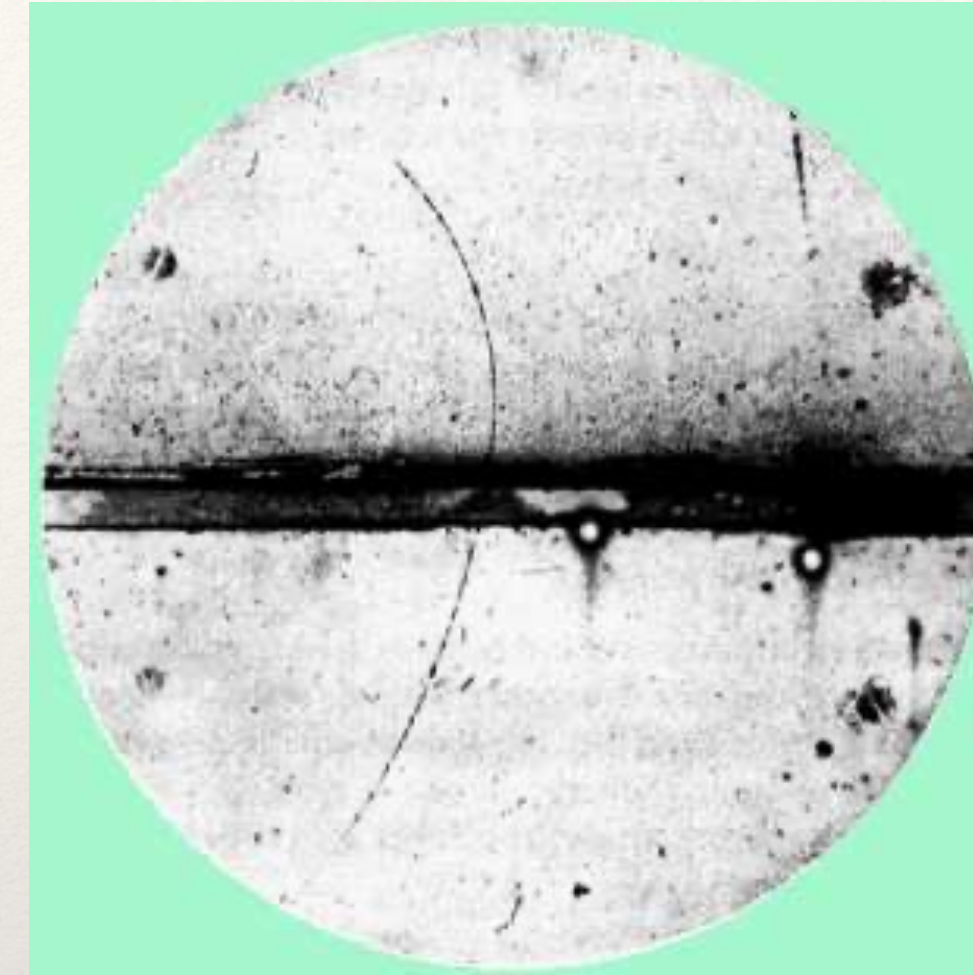
Hydrogen atom and spectra can be explained by Coulomb potential

Relativistic Quantum Mechanics and quantum fields

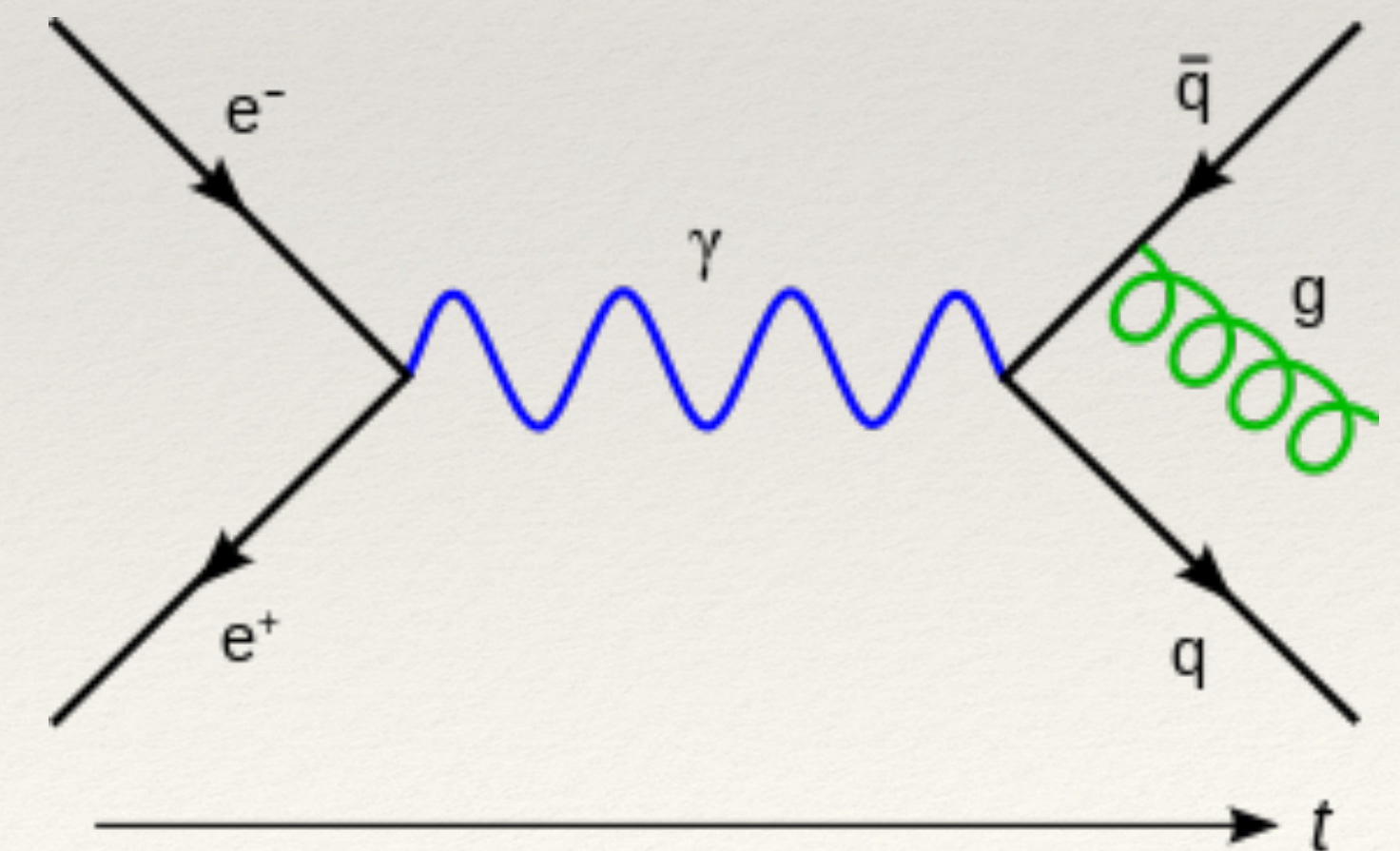
- ❖ A triumph of relativistic formulation of Quantum mechanics, which started with Klein Gordon Equation and leads to Dirac Equation showed that each particle has Antiparticle ! Like positron for electron.

However, creation and annihilation of particles could only be described by Quantum field theory, where particles are assumed as quantised fields. Moreover,

- * The interactions between the fundamental particles are carried by particles
- * Electromagnetic interaction: Photon
- * Weak interaction: W and Z boson
- * Strong interaction: Gluon
- * Gravitational interaction: Graviton (not discovered yet)



Discovery of
positron in
1932





LARGE HADRON COLLIDER (LHC)

Started running in 2008 at CERN,
Geneva in Switzerland

One of the biggest experiments
ever built to probe the quantum
nature of the universe !



July 4, 2012 : Discovery of Higgs Boson !

A particle that generates mass for all
the other fundamental particles

One of the biggest discoveries of this decade

LHC is now looking for:

**Dark Matter,
a big mystery !**

"As long as people were satisfied that an apple falls to the ground because that is its natural place, even the basic properties of motion remained hidden. A willingness to be puzzled by the simplest phenomena is the very beginning of science."

Noam Chomsky, "Language and thought"

Thank you !

